

A Task-Routed Hybrid Vision–Language System for Medical Visual Question Answering

and a Calibrated Analysis of the Open-Ended Reasoning Ceiling

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1 Motivation & Benchmark

MedReason benchmarks medical multimodal reasoning: **MCQ** + **open-ended** questions, each needing an image-grounded trace.

- Clinical reliability needs correct, grounded answers.
- Confident wrong findings are the core risk.
- Tension**: fine-tuning lifts MCQ (0.33→0.945) but shortens traces (44→20 words).

17,722 cases 14,398 MCQ 3,324 open

Scored on: **MCQ acc** · **GT agreement** · **VA** grounding

2 Method: Task-Routed Hybrid

One **Qwen2.5-VL-7B** backbone + one LoRA adapter (~40 MB), **toggled per task**: on for MCQ, off for open-ended. ~40 MB adapter replaces a second full model.

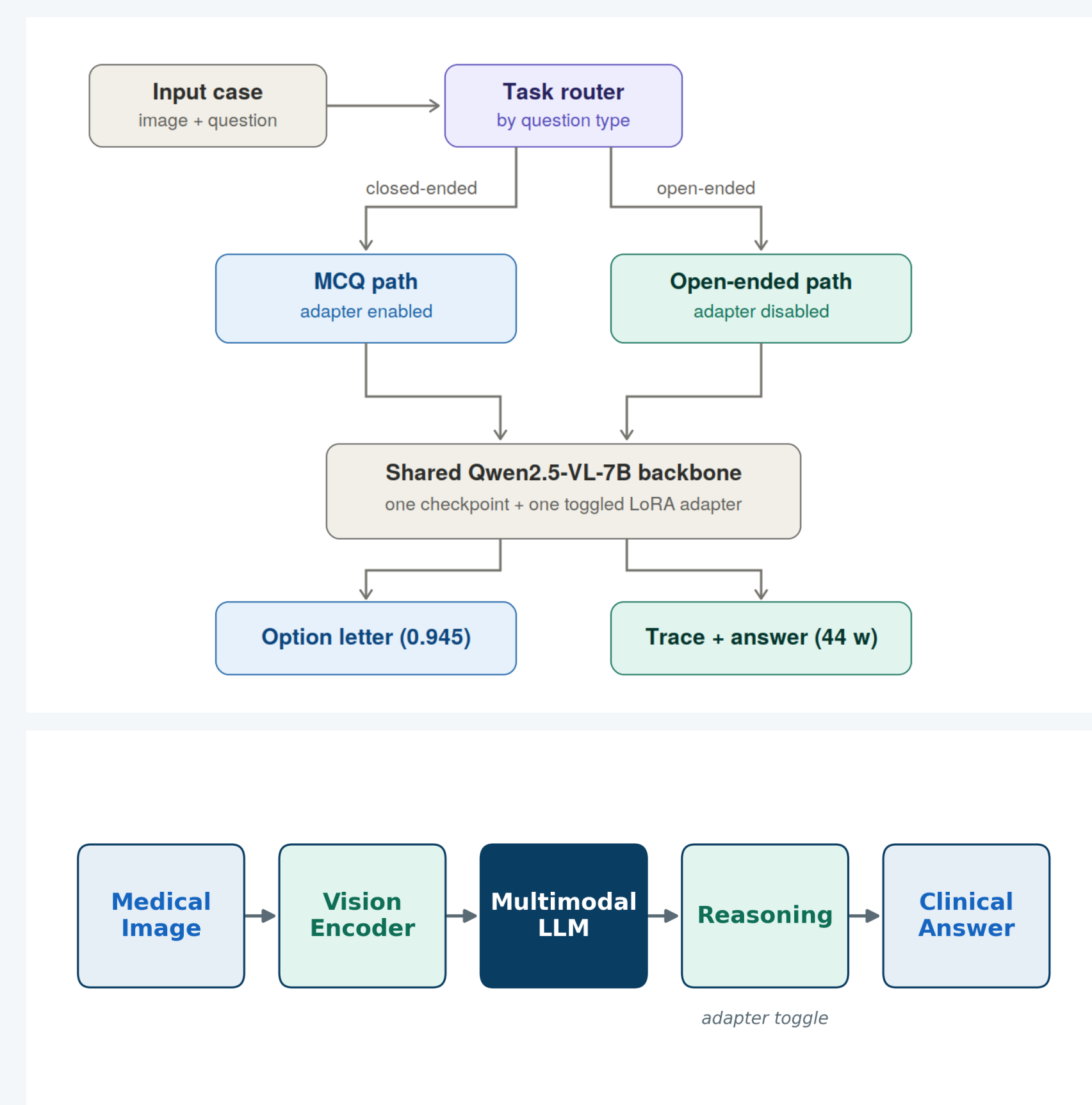


Fig. 1. Adapter ON for MCQ, OFF for open-ended; one shared backbone.

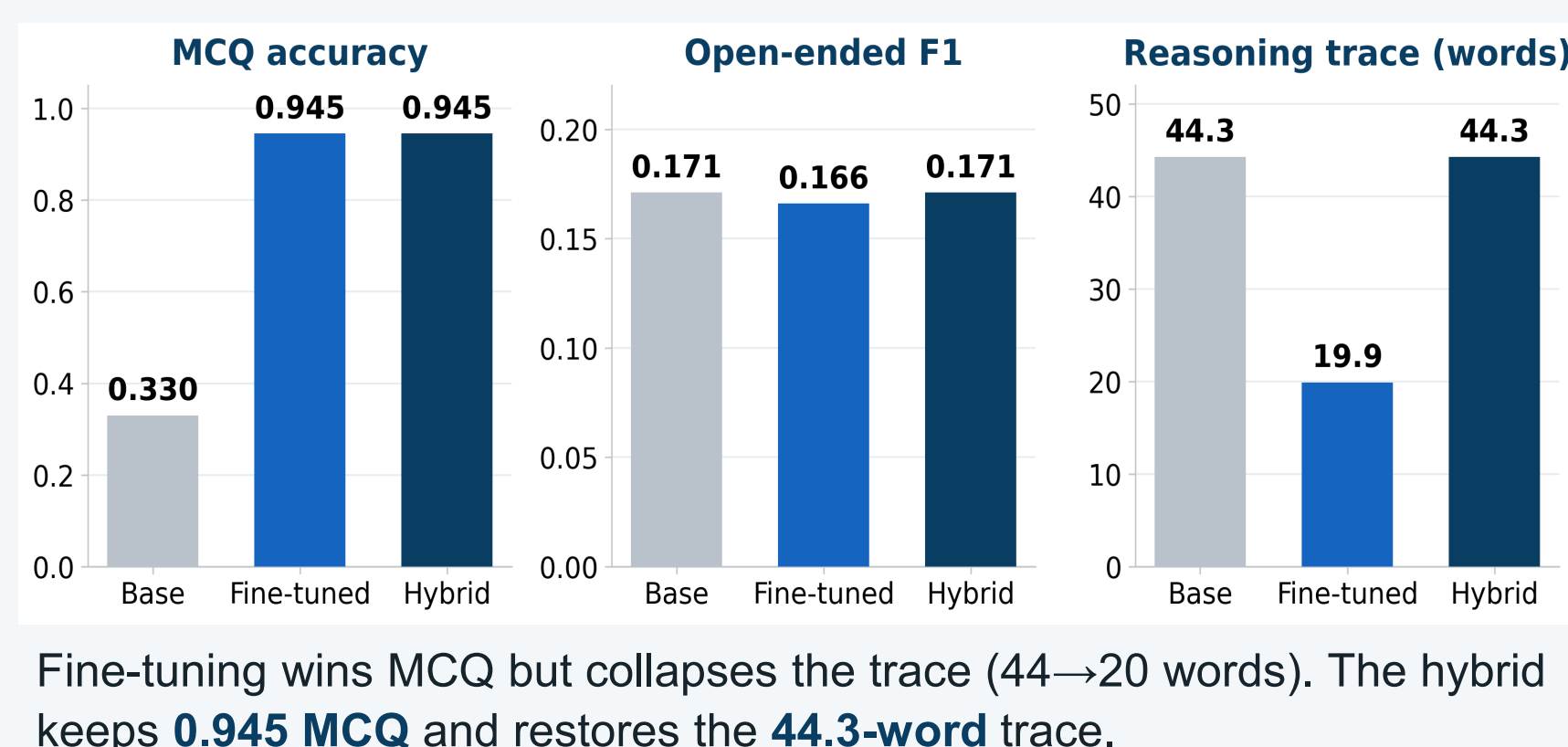
3 Why It Works

MCQ training rewards single-letter answers, and that brevity leaks into open-ended generation.

Turning the adapter OFF restores the base model's fuller, image-grounded traces, **recovering GT and VA without losing MCQ accuracy**.

This is why routing beats a single compromise model: each task gets the configuration that serves it best, from one backbone.

4 Key Result: Base vs Fine-tuned vs Hybrid



5 Calibrated Local Judge

Overlap metrics can't separate systems (all within 0.016 F1). A local **Qwen2.5-32B judge** (0–4) **reverses a false token-F1 “win”**: the open-only LoRA scores worse, not better.

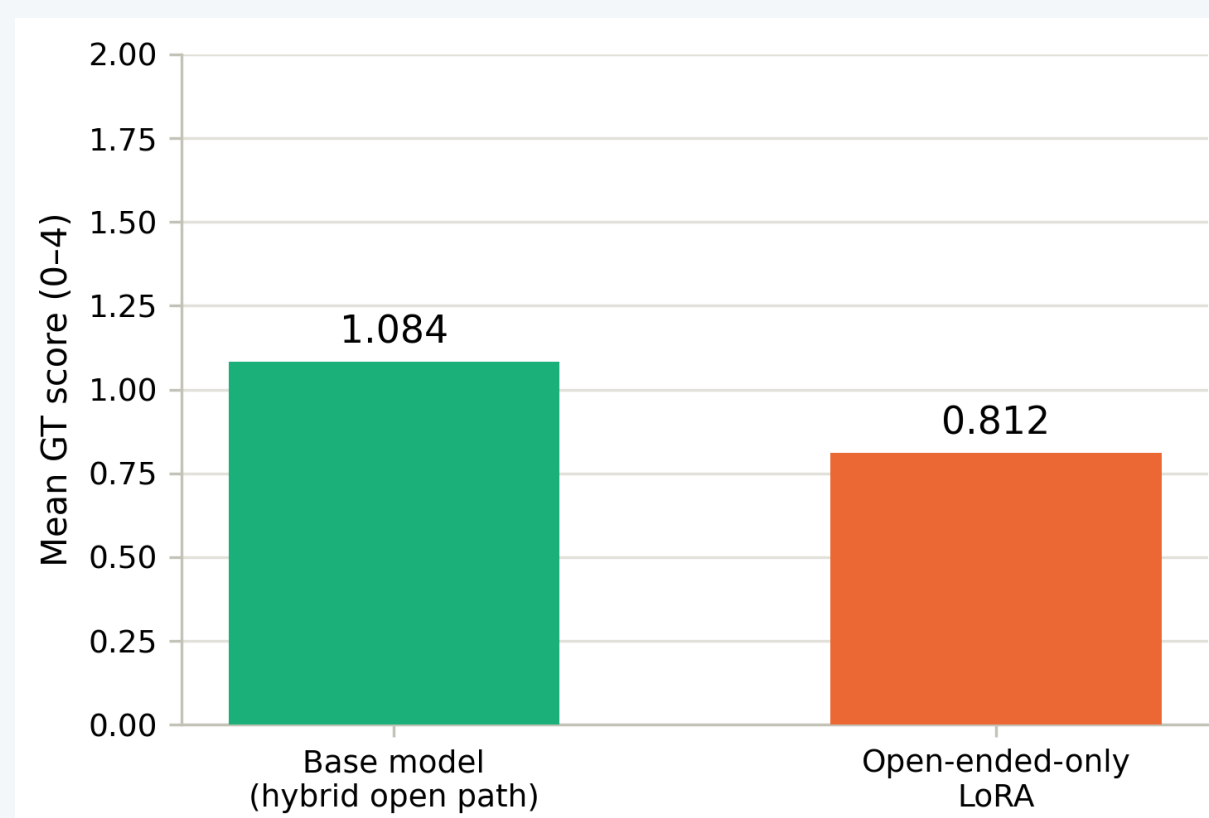


Fig. 2. Base model beats open-only LoRA (GT 1.084 vs 0.812).

6 What Did Not Work

✗ **Self-consistency**: vote@5 = greedy = 0.945 ($\Delta +0.000$), 6x cost.

✗ **DINOv3 retrieval**: apparent F1 0.285 was self-leakage (sim 1.000); leakage-free scored 0.159, below baseline.

✗ **Prompt variants**: no variant beat the existing prompt.

✗ **Open-only LoRA**: token-F1 said better (0.175); the judge said worse (0.812 vs 1.084 GT).

7 A Perceptual, Not Expressive, Ceiling

On the hardest cases the model **confidently names the WRONG finding**.

“right coronary cusp” → “**myocardial infarction**”
“eosinophils” → “**atypical lymphocytes**”

These are failures to **perceive** subtle findings, not to express them. Output-side fixes (prompting, retrieval, adaptation) cannot move a visual bottleneck.

8 Where It Fails

By question type, the hardest are those needing a **specific finding**: diagnosis, location, open reasoning.

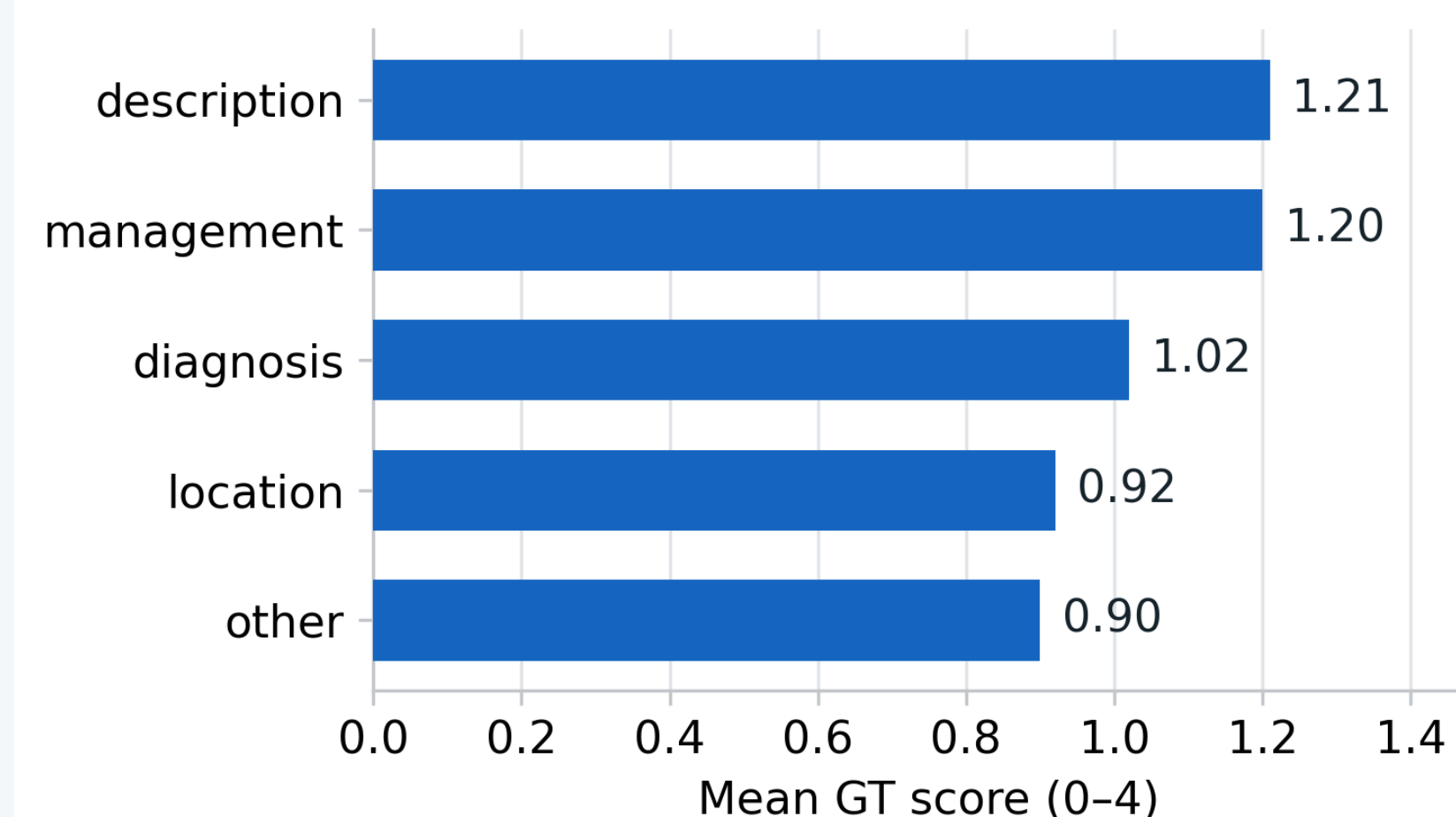


Fig. 3. Hardest types need a specific finding (mean GT, 0–4).

9 Limitations & Future Work

- Judge anchored to one baseline point; needs human or official-judge agreement.
- Cause of terseness not fully isolated (adaptation vs. objective).
- Next**: stronger / domain-pretrained vision backbone; image-removal & question-only controls.

SCAN FOR
PROJECT
RESOURCES

GitHub · Paper · Challenge ·
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